CE 766: Watershed Management · Project Proposal

Robust phosphorus-loading policies for Powai Lake, Mumbai: reformulating the Lake Problem under deep uncertainty (2017–2025)

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1. Problem definition

Powai Lake (Mumbai, India) is eutrophic. Exact problems: (1) about 18 million litres/day of untreated sewage has loaded phosphorus into a 2.1 km² lake, causing water-hyacinth cover, algal blooms and fish kills; (2) phosphorus stored in sediments can recycle, so the lake may stay eutrophic even after sewage is diverted; (3) monsoon runoff adds an uncontrolled, highly variable load; (4) the recycling rate, removal rate and critical threshold are unknown for this lake. Decision question: what annual phosphorus load should be permitted during 2026–2050 so that pollution-control cost is minimised while the lake reliably stays below the eutrophication threshold?

(99 words)

2. Three main intended outcomes

1. Problem structuring: a Powai-specific lake model in XLRM form, with the recycling rate, removal rate and natural-inflow statistics estimated (as plausible ranges, not single values) from rainfall records, published water-quality measurements and a satellite chlorophyll proxy.
2. Identifying solutions: a Pareto-approximate set of phosphorus-loading policies that exposes the trade-offs among pollution-control cost, peak lake phosphorus, reliability of staying below the critical threshold, and year-to-year policy stability.
3. Inferred relationships: the combinations of uncertain factors under which policies fail (scenario discovery), leading to a robust recommendation on whether the planned diversion and 8 MLD treatment plant are sufficient or deeper load cuts are needed.

3. Study area

Powai Lake (about 19.13°N, 72.91°E) is an artificial lake in the northern suburbs of Mumbai, Maharashtra, India, created in 1891 by damming a tributary of the Mithi River, and it adjoins the IIT Bombay campus. It has a water spread of roughly 2.1–2.2 km² and a small catchment of about 6.6 km² in the Powai–Kanheri hills, now heavily urbanised. Originally 3–12 m deep, it has become much shallower through siltation. The region receives more than 2,000 mm of rain a year, almost all in June–September, so natural nutrient inflows are strongly seasonal. The water is unfit for drinking and is classed as eutrophic.

Analysis period: 2017–2025 for lake-state observations (Sentinel-2) and 1991–2025 for rainfall; policies are simulated over a 25-year planning horizon.

4. Methodology

1. Framing (XLRM): lever = annual anthropogenic phosphorus load allowed into the lake (degree of sewage diversion/treatment); uncertainties = recycling exponent q, removal rate b, mean and variance of natural inflows, discount rate.
2. Data: IMD gridded daily rainfall (1991–2025) converted to runoff with the SCS curve number method to generate lognormal natural phosphorus inflows; Sentinel-2 NDCI chlorophyll proxy (2017–2025, Google Earth Engine) and published water-quality values to bound the lake state and parameters.
3. Model: Carpenter et al. (1999) lake phosphorus balance, coded in Python.
4. Optimisation: direct policy search with radial-basis-function rules (Quinn et al., 2017) using NSGA-II over 100 stochastic replicates and a 25-year horizon. Objectives: maximise economic benefit (avoided treatment cost), minimise peak phosphorus, maximise reliability of staying below the critical threshold, maximise policy inertia.
5. Robustness: re-simulate Pareto policies across 1,000 Latin-hypercube states of the world, compute satisficing robustness, and use PRIM scenario discovery to find failure conditions; compare against the planned 8 MLD plant as baseline.

(157 words)

5. Supporting information: references and data sources

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11. Data: IMD gridded rainfall (https://imdweb.lwcc.in/); Copernicus Sentinel-2 surface reflectance via Google Earth Engine; Brihanmumbai Municipal Corporation releases on Powai Lake sewage diversion and treatment (2025–2026).